Elective 4 (Information System Development and Management)

LABORATORY ACTIVITY #2

IT 415

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EXPECTED OUTCOMES

- This activity will deepen students' understanding of software engineering concepts by applying real-world practices and frameworks, culminating in a robust information system project.

- The activity will also foster teamwork, communication, and critical thinking skills.

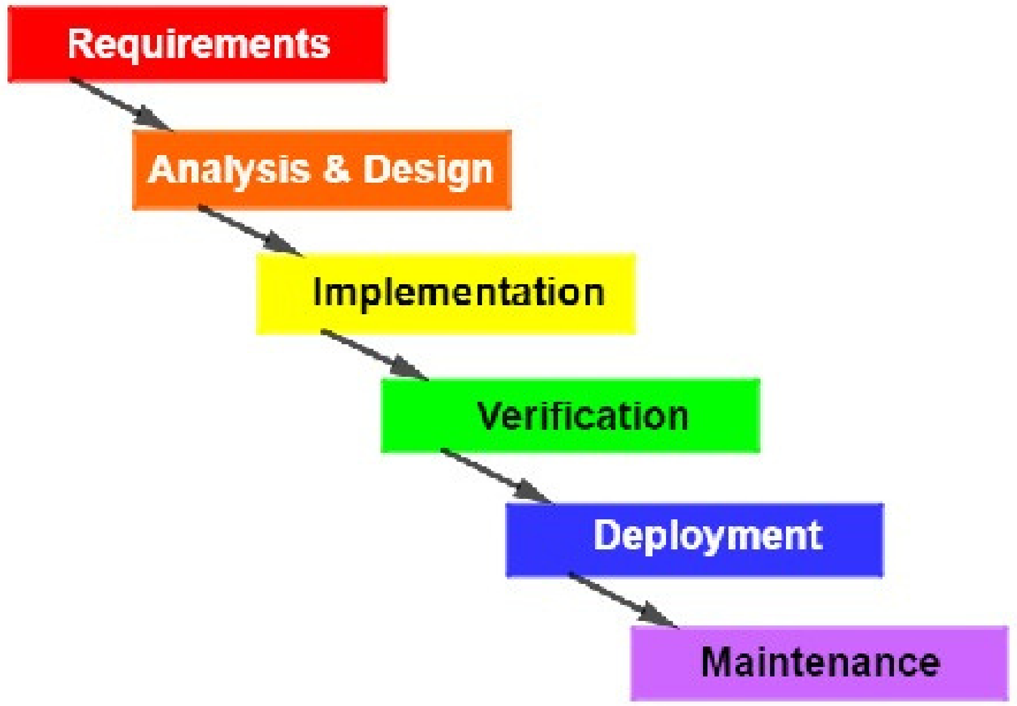
Continuation of Laboratory Activity #1:
Application of Software Processes, Models, and Quality Factors

Objective:

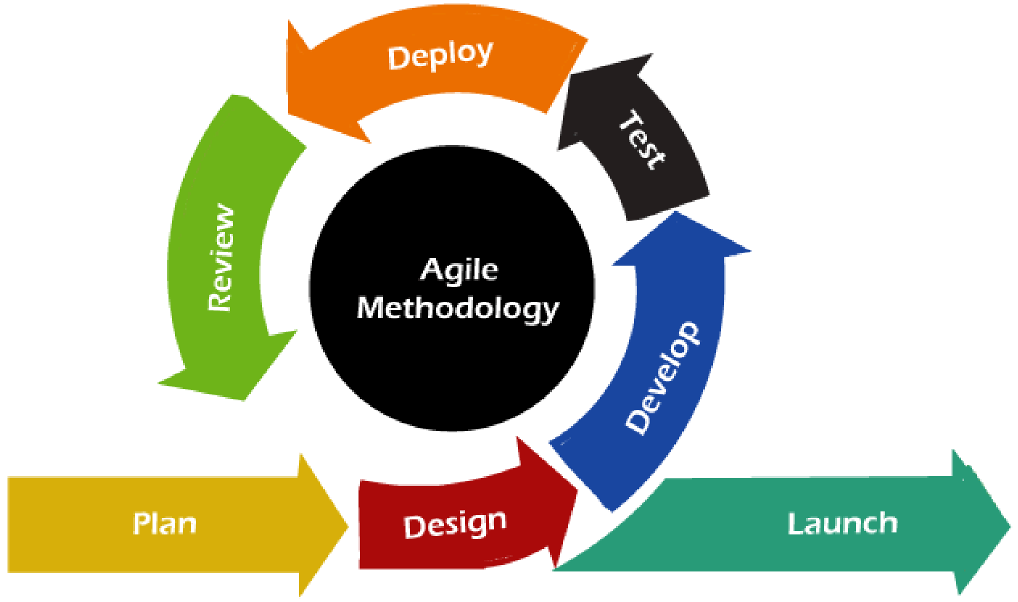
- Deepen students' understanding of software engineering by applying sof tware process models and McCall's Software Quality Factors. Students will learn to select, justify, and implement these models, ensuring quality in a real-world project while enhancing their practical skills in softwar development and teamwork.

Software Process Models

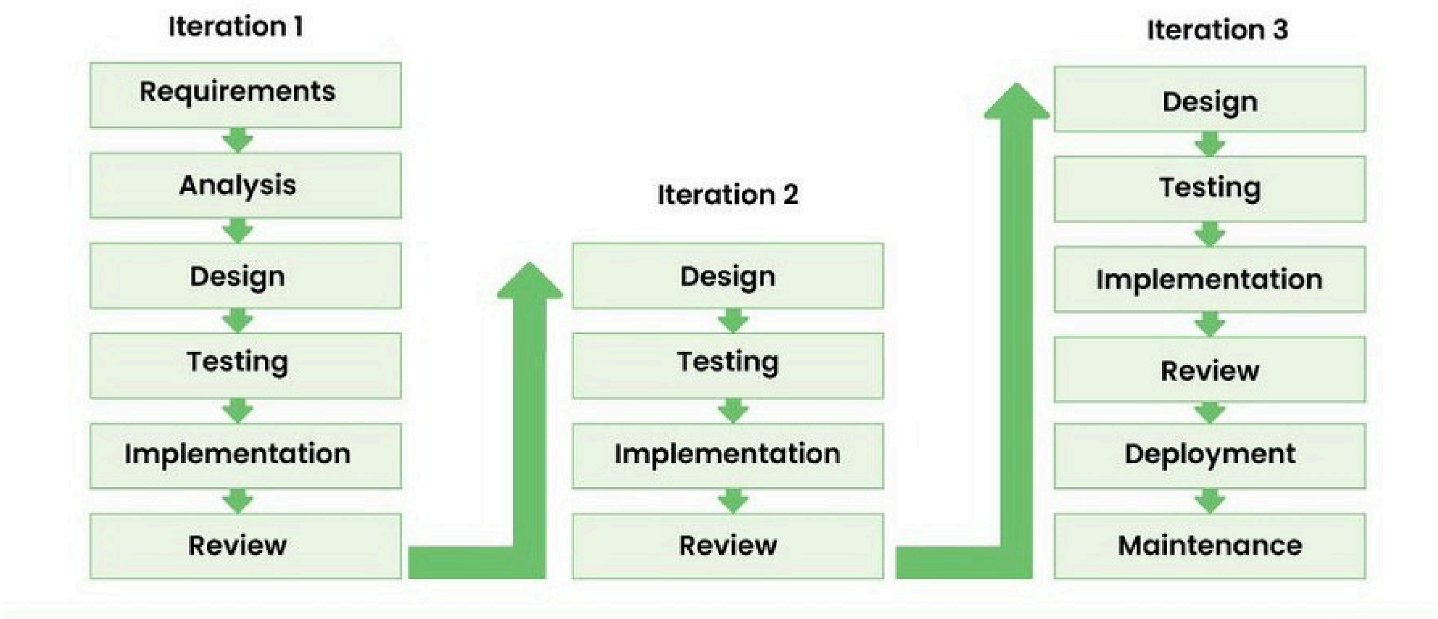
The Waterfall Model



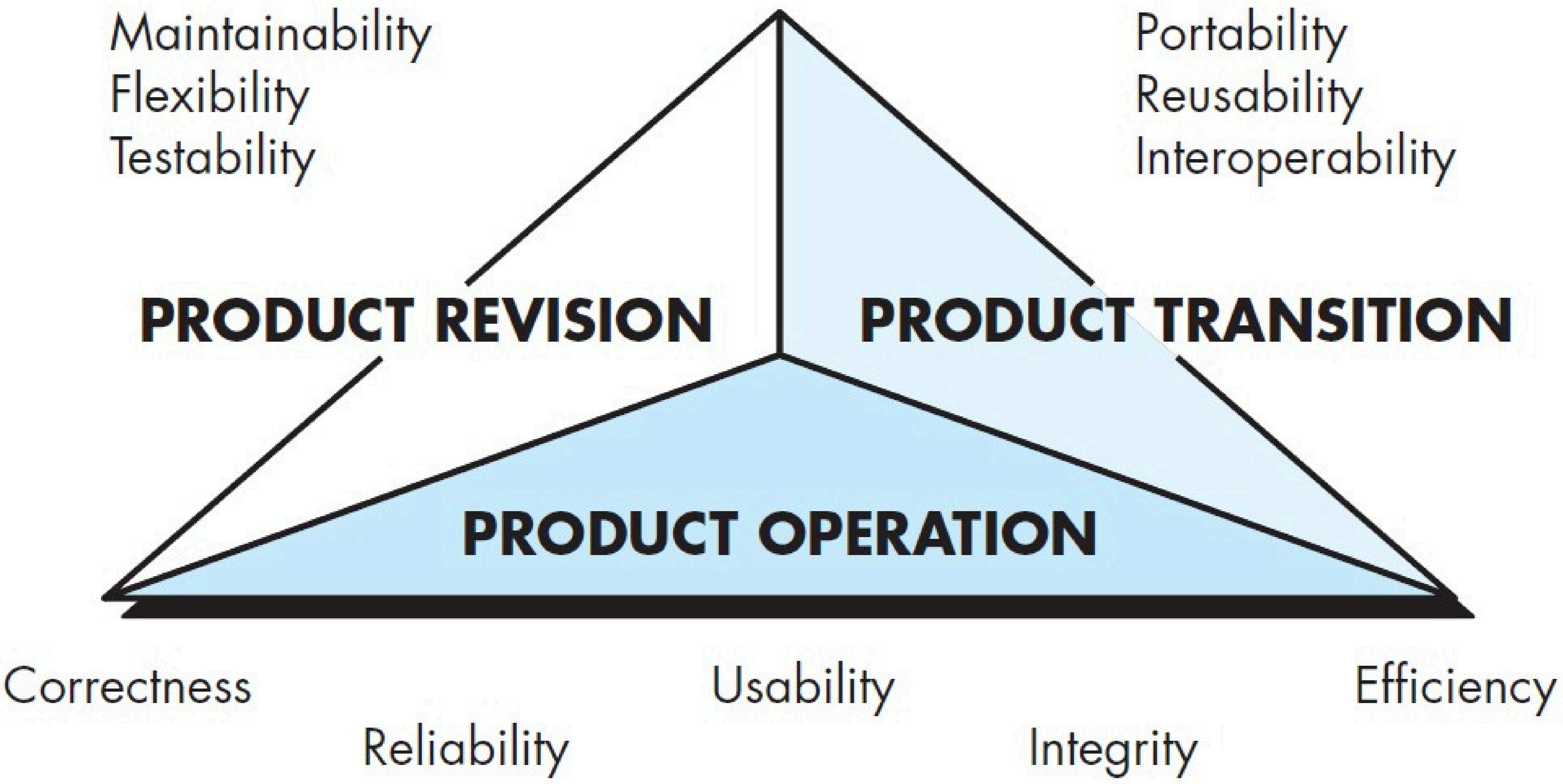
Agile Model



Iterative Model



McCall's Software Quality Factors Model



Continuation of Laboratory Activity #1:
Application of Software Processes, Models, and Quality Factors

Step 1: Apply a Software Process Model

- **Objective:** Based on the project selected and the SDLC process applied, each group should select and implement a specific software process model.

Instructions:

- Research and choose an appropriate software process model that aligns with your project's requirements.
- Justify the selection of the model, considering factors like project scope, complexity, and team dynamics.
- Document how each phase of your selected software process model will be implemented within your project.

Continuation of Laboratory Activity #1:
Application of Software Processes, Models, and Quality Factors

Step 2: Implement McCall's Software Quality Factors

- **Objective:** Assess and ensure that your project meets McCall's Software Quality Factors Model criteria.
- **Instructions:**
  - Identify and explain the relevance of McCall's quality factors (e.g., reliability, maintainability, usability, efficiency, etc.) in the context of your project.
  - Develop a strategy to incorporate these quality factors into your system's design and implementation.
  - Provide a plan for how you will measure and validate the quality factors throughout the development process.

Continuation of Laboratory Activity #1: Application of Software Processes, Models, and Quality Factors

Step 3: Final Presentation and Quality Assessment

- **Objective:** Present the final system design, emphasizing how the chosen software process model and McCall's quality factors were applied.
- **Instructions:**
  - Prepare a final presentation that includes an overview of your project, the selected software process model, and how McCall's software quality factors were integrated.
  - Highlight any challenges encountered and how they were addressed using the process model and quality factors.
  - Discuss the results of your quality assessments and peer reviews, and how they influenced your final design.